Wesley Jennings

WesleyJennings.com | Wildwood, MO | (314) 939-8792 | Wesjennings04@gmail.com

Education

University of Missouri Columbia, MO
B.S. Computer Science, Mathematics Minor May 2026
Master's of Computer Science Expected May 2027
Cumulative GPA: 3.85/4.0
Coursework: Algorithm Design 1-3, Parallel Programming (HPC), Operating Systems, Computer Networks, Object-Oriented Programming, Calculus I - III, Differential Equations, Machine Learning, Software Security, Artificial Intelligence, Computer Org and Assembly Language, Database Application and Info Systems

Skills & Interests

Languages: C, Java, Python, C++, Swift, JavaScript, Bash
Frameworks: Spring, HTML, CSS
Databases: SQL, Oracle, PostgreSQL
Tools & Libraries: Git

Experience

NISC, Backend Software Developer Intern May 2026 - Aug 2026

- Replaced a legacy Crystal Reports batch job with a Spring Boot / Spring Batch application, using Hibernate ScrollableResults to stream across five joined entities for memory-efficient processing on both Oracle and PostgreSQL.
- Built a multi-section PDF reporting engine with Apache PDFBox (dynamic grouping, subtotals, grand-total aggregation, cross-category deduplication), slated for the company's annual year-end processing.
- Refactored a legacy Java billing module with polymorphic enums and consolidated SQL, cutting code volume ~40% with verified behavioral parity, and implemented configurable provider-priority ordering for credit transfers.
- Verified cross-engine correctness with JUnit 5 and Testcontainers integration tests, and automated credit-transfer batch runs with a bash script integrated into the company's year-end processing pipeline.

Projects

Unix-Style Filesystem, C Oct 2025 - Dec 2025

- Implemented an inode-based filesystem from scratch in C over a 256 MB virtual block store, supporting format, mount, unmount, and persistent storage across sessions.
- Designed a three-level block addressing scheme (6 direct pointers, single indirect, and double indirect) enabling large-file support with on-demand block allocation and full read/write/seek semantics tracked per file descriptor.
- Built a POSIX-like API (create, open, close, remove, move, link, get_dir) with hard links via inode reference counting, absolute-path directory traversal, and a bitmap-managed free inode table.

NBA Spread Prediction Model, Python Sep 2025 - April 2026

- Built a multi-model NBA spread analysis pipeline combining four versions of a statistical edge model, including L1 and L2 logistic regression, to compute projected margins, fair spreads, cover probabilities, and per-game expected value.
- Integrated GPT-4.1-mini as a synthesis layer with a structured system prompt and enforced JSON schema output, feeding it the quantitative signals to generate a final pick, confidence score, and risk flags for each game.
- Designed an end-to-end pipeline pulling historical game data from a Neon PostgreSQL database and live odds from the NBA API, with a tkinter GUI providing live and demo prediction modes with progress tracking and result export.

Homemade Beowulf Cluster, Linux/C Feb 2026 - April 2026

- Built a commodity Beowulf cluster from old desktops and laptops, installing Ubuntu Server and configuring static IP networking, hostname resolution, and passwordless SSH across nodes.
- Configured an NFS-shared filesystem and a common MPI user across the cluster, and deployed MPICH with the Hydra process manager to distribute jobs across nodes.
- Validated the cluster by creating and running MPI test programs across all nodes, verifying process distribution and inter-node communication through MPICH.